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Pot Experiments Overestimate Mercury Accumulation in Rice: Evidence from Multi-Year Field Validation

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Abstract

The uptake and accumulation of mercury (Hg) in rice poses a serious threat to food safety. Pot experiments are widely used to screen for low-Hg-accumulating cultivars, yet their reliability in predicting field performance remains uncertain. This study evaluated pot-based screening by (1) comparing Hg uptake in rice grown in freshly processed versus aged soil; (2) contrasting Hg accumulation in the same cultivars grown in pots versus at two field sites; and (3) isolating micro-environmental effects by burying pots in situ. A total of 22 rice cultivars were used during 2021–2023 in this study. Pot systems, regardless of soil treatment, failed to replicate field accumulation patterns, yielding significantly greater Hg concentrations in brown rice (up to 59.24 ng g^{-1}) than field conditions (maximum 32.33 ng g^{-1}). Cultivar rankings derived from pot experiments showed little or no correlation with field rankings, indicating that performance is not transferable across environments. Random forest analysis identified elevated soil temperature and reduced light intensity as key artificial factors driving overestimation in pots, explaining 15.68% (total Hg) and 21.65% (methylmercury) of the variation. We conclude that pot experiments—due to soil disturbance and altered microclimates—overestimate Hg accumulation potential and show limited predictive capacity under the tested conditions. Therefore, field validation across multiple sites and seasons is essential for accurate mercury risk assessment and region-specific cultivar recommendation.

Keywords: mercury accumulation; rice paddy; pot experiment; field validation; cultivar variation

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1. Introduction

Rice (*Oryza sativa* L.) is a staple food for over half of the world's population, with milled rice production projected to account for approximately 18.6% of global cereal output by 2025 [1]. However, rice paddies are recognized as a significant hotspot for mercury (Hg) bioavailability. During growth, rice plants efficiently absorb Hg, particularly its neurotoxic organic form, methylmercury (MeHg), from soil and the atmosphere, leading to substantial enrichment in grains [2,3]. In inland regions of China, rice plants replace fish as the primary source of methylmercury exposure [4,5], posing serious threats to human neurological, cardiovascular, and renal health [6–9].

The translocation, transformation, and uptake of mercury within the soil–rice system are regulated by a complex interplay of physical, chemical, and biological factors. Specifically, agricultural management practices like straw returning can enhance mercury methylation in soil [10], whereas drought, and alternate wetting and drying treatments can significantly reduce soil and grain MeHg concentrations [11,12]. These processes are largely driven by soil microorganisms. For instance, iron-reducing and sulfate-reducing bacteria can produce methylmercury in soil [2,13,14], while anaerobic microorganisms such as *Pseudarthrobacter* (genus), *Dechloromonas* (genus), and *Methylophilaceae* (family) have been reported to participate in MeHg degradation [15]. Two main mitigation strategies, i.e., soil amendments [16,17] and the screening of low-Hg-accumulating rice cultivars [18] have been developed.

Due to their high controllability, pot experiments are widely employed to study the effects of these complex factors on Hg accumulation in rice. The total mercury and methylmercury bioconcentration factors (BCFs) in pot experiments vary greatly [19]. However, pot environments represent a highly simplified simulation of field ecosystems. Small pots restrict root growth volume [20], disrupt the microbial community structure of intact soil [21], and are prone to creating more anoxic conditions [22], potentially amplifying experimental biases. Fundamental differences between the two scales—in terms of soil structure, water dynamics, microbial communities, root development space, and material cycling—mean that results from pot experiments may not fully, and could even erroneously, predict field outcomes. To our knowledge, no previous study has systematically compared pot, buried-pot, and multi-site field systems across multiple years.

To fill this gap, this study aimed to systematically evaluate the accuracy of pot-based experiments in predicting mercury (Hg) accumulation in rice grains under field conditions. Specifically, we sought to: (1) compare Hg uptake in rice grown in freshly processed versus aged pot soil; (2) contrast Hg accumulation in the same cultivars grown in pots versus at two field sites; and (3) isolate microenvironmental effects by burying pots in situ. We hypothesize that: (1) the artificial microclimate in pot systems—characterized by elevated soil temperature and attenuated light intensity—overestimates grain Hg accumulation compared to field conditions; (2) cultivar performance in Hg accumulation is context-dependent, with no consistent ranking across different field environments, necessitating region-specific validation.

2. Materials and Methods

2.1. Experimental Design

This study was conducted over three consecutive years (2021–2023). Pot experiments were carried out at the Jiaxing Academy of Agricultural Sciences, Zhejiang Province, China (120.7164, 30.8322), while field experiments were conducted at two sites in the Yangtze River Delta: Field A in Zhejiang Province (120.8220, 30.7617) and Field B in Jiangsu Province (120.5078, 31.4704). The experimental design comprised three complementary components: (1) a pot experiment comparing freshly processed versus aged soil to examine soil history effects; (2) a parallel pot and field experiment using the same rice cultivars to assess the transferability of cultivar rankings; and (3) a buried-pot experiment to isolate microenvironmental effects on Hg accumulation. Detailed descriptions of each component are provided below.

2.1.1. Pot Experiment

The pot experiments were conducted over three consecutive years (2021–2023) to screen for low-Hg-accumulating rice cultivars and to investigate microenvironmental effects on Hg accumulation. The soil used for potting was collected from a Hg-contaminated paddy

field in Southern Jiangsu. The basic physicochemical properties of the soil were as follows: pH 4.76, organic matter 47.4 g kg^{-1}, total mercury (THg) 0.75 mg kg^{-1}, available nitrogen 223 mg kg^{-1}, available phosphorus 14.2 mg kg^{-1}, and available potassium 289 mg kg^{-1}. The pots used are all ceramic pots with a diameter of 20 cm and a height of 24 cm. Before potting, the water content of the stored soil was measured, and 5 kg of dry soil was filled into each pot at a bulk density of 1.2 g cm^{-3}.

In 2021, eleven cultivars were tested; in 2022, nineteen cultivars were used, including all the cultivars that were used in the 2021 experiment (Table S1). The tested cultivars were selected based on their widespread cultivation in the Yangtze River Delta region, covering both conventional and hybrid rice types, to ensure the representativeness of the findings within the study area.

For the 2022 pot experiment, after the 2021 harvest, all residual roots were removed from the pots, and the soil was air-dried naturally for at least six months during the non-cropping period. The soil was then flooded for two weeks before transplanting on July 14. Total Hg concentrations measured after the 2021 and 2022 pot trials were 0.70 mg kg^{-1} and 0.71 mg kg^{-1}, respectively.

In 2023, a pot experiment was conducted using three rice cultivars (Nanjing 3908, Nanjing 5055, and Jia 58, all *Oryza sativa* subsp. *japonica* Kato) to assess microenvironmental effects on Hg accumulation. The experiment consisted of indoor and outdoor components: the indoor treatment was Greenhouse Pot (GP), comprising conventional pots placed inside a glass greenhouse. The outdoor treatments were established in an unpolluted experimental field (pH: 6.30 $\pm$ 0.06, THg: 0.23 $\pm$ 0.01 mg kg^{-1}) and included three designs: Buried Pot (BP), where pots placed in pits dug to field level with the soil surface level with the field to simulate field root-zone temperature (pots were laterally and vertically impermeable, and the rim was raised by 6 cm to prevent field flooding from interfering with the alternate wetting and drying cycle applied to all treatments); Block-flowing PVC Ring (BR), where PVC rings of the same inner diameter as pots were vertically placed into pits, filled with the same soil as GP, allowing vertical but not lateral water and air exchange; and Clean Soil (CS), where rice seedlings were transplanted directly into the unpolluted soil to exclude the influence of the test soil on brown rice Hg accumulation.

The agronomic parameters (e.g., growth period, plant height, yield) for the 2023 pot treatment are summarized in Table S2. To isolate the cultivation environment as an experimental variable, atmospheric deposition was monitored monthly at three points within and near the buried-pot plots (sampler locations and field setup are shown in Figure S1; deposition data are provided in Table S3). Detailed methods for dry and wet deposition sampling are described in the Supplementary Information. For the potted plants in the glass greenhouse (GP), soil temperature (°C), air temperature (°C), light intensity (Lux), and relative humidity (%) were recorded using a remote temperature and humidity data logger (UbiBot, GS1-AL4G1RS, China). Parallel recordings for BP, BR, and CS were performed by equipping data loggers of the same model with additional temperature and humidity probes.

For all pot experiments, treatments were arranged in a randomized complete block design, with each treatment having three replicates (one pot per replicate). Each pot contained three hills, with one plant per hill. All pots were repositioned twice during the growing period. At harvest, the three plants from the same pot were pooled as a single biological replicate for sample analysis. During the rice growth period, a water-soluble fertilizer containing $CO(NH_2)_2$, KH_2PO_4, and K_2SO_4 was applied via hole placement. Application rates followed optimal N–P–K ratios for rice: 200 kg ha^{-1} for conventional rice and 250–300 kg ha^{-1} for hybrid rice. The specific formulation and dosage are described

in Table S4. Essential procedures and key dates for rice cultivation are provided in the Supplementary Information.

2.1.2. Field Experiments

Two field experiments were conducted in 2022 to provide reference data for Hg accumulation under real paddy conditions. Each treatment had three replicates, with each replicate consisting of one plot (21 m^2 for Field A and 10 m^2 for Field B). Soil properties for Field A were: pH 5.82, organic matter 33.1 g kg^{-1}, and THg 0.33 mg kg^{-1}; for Field B: pH 6.41, organic matter 38.58 g kg^{-1}, and THg 0.93 mg kg^{-1}. Field water management followed local farmers' practice, during which the field was drained for approximately 7 days at the late tillering stage, then re-flooded with a water layer of more than 5 cm, with irrigation terminated two weeks before harvest.

For Field A, the same 11 rice cultivars were used as in the pot experiments. For Field B, the cultivar set was identical to that used in the pot cultivar screening experiments.

In each field plot, the herbicide Pretilachlor (50% active ingredient) was applied for pre-emergence weed control one week before flooding. The field was then flooded and leveled, and seedlings were transplanted 3–5 days after flooding. Treatments were arranged in a randomized complete block design. These field experiments provided reference data for Hg accumulation under real paddy conditions.

2.2. Sample Collection

Field soil samples were collected using a five-point sampling method and mixed. At crop harvest, fresh soil samples (0–20 cm) were collected using a soil auger, placed in polyethylene zip-lock bags. Simultaneously, approximately 1 kg of mature rice plants was collected and placed in nylon mesh bags for transport to the laboratory.

Pot soil samples were mixed from three points. At harvest, fresh soil samples were collected using a small-bore pot soil auger (inner diameter 16 mm, height 500 mm) and placed in polyethylene bags. Simultaneously, all rice panicles (including effective and ineffective ones) within the pot were collected and placed in nylon mesh bags.

Soil samples were freeze-dried to constant weight, quickly broken into small pieces with a wooden hammer, and quartered to retain half. This portion was rapidly ground with a mortar and passed through a 2 mm nylon sieve. It was quartered again, and half was finely ground and passed through a 0.149 mm nylon sieve. The processed soil was stored in polyethylene zip-lock bags at −80 °C to preserve methylmercury stability.

Plant samples were oven-dried below 50 °C to constant weight, then threshed and dehusked. Brown rice was ground and passed through a 0.149 mm nylon sieve, then stored in polyethylene zip-lock bags for later use.

2.3. Sample Analysis and Quality Control

Soil pH was measured using a soil-to-water ratio of 1:2.5 by pH meter (S220, Mettler Toledo, Schwerzenbach, Switzerland). Organic matter concentration was determined by potassium dichromate-sulfuric acid oxidation, titrated with ferrous sulfate standard solution with organic carbon concentration multiplied by the constant 1.724. Total nitrogen was analyzed using a fully automatic Kjeldahl nitrogen analyzer (K9840, Haineng, Jinan, China). Cation exchange capacity (CEC) was determined using the trichloride hexaamminecobaltate extraction-spectrophotometric method (UV-2700, Shimadzu, Kyoto, Japan). Soil available N (Alkali-hydrolyzable N) was determined after hydrolysis with sodium hydroxide solution, followed by titration with hydrochloric acid standard solution after adding boric acid-indicator solution. Available phosphorus was extracted with ammonium fluoride-hydrochloric acid solution and determined by molybdenum-antimony anti-colorimetry (UV-2700, Shimadzu, Kyoto, Japan). Available potassium was measured

using a flame photometer (6400A, Shanghai Precision Science, Shanghai, China), with GBW07408a (GSS-8a, Institute of Geophysical and Geochemical Exploration, Langfang, China) used as the standard reference material (recovery rate: 98.0% to 101.9%). Soil total mercury was determined using catalytic pyrolysis-cold atomic absorption spectrophotometry (DMA-80, Milestone, Sorisole, Italy). For soil methylmercury, samples were digested and extracted with 25% KOH-methanol solution and quantitatively determined by gas chromatography-cold vapor atomic fluorescence spectrometry (MMA72, Polytech, Beijing, China), with ERM-CC580 (JRC, Belgium) used as the standard reference material (recovery rate: 82.8% to 91.4%). Quartz sand was used as a blank.

Plant samples were digested with HNO_3 using a microwave digestion system. Total mercury content was analyzed by atomic fluorescence spectrophotometry (BAF-2000, Baode, Beijing, China), with GBW10015 (GSB-6, Institute of Geophysical and Geochemical Exploration, Langfang, China) used as the standard reference material (recovery rate: 90 to 107%). For methylmercury analysis, samples were digested with 30% nitric acid at 60 °C and quantified by gas chromatography–cold vapor atomic fluorescence spectrometry (MMA72, Polytech, Beijing, China). Matrix spike recoveries were used for quality control. Quality control included at least 10% parallel samples, blank spikes, and sample spikes for all analyses (see Table S5 for specific detection limits and detailed recovery data). Detailed methods for methylmercury analysis are provided in the Supplementary Information.

2.4. Data Calculation and Statistics

To exclude the influence of the soil background value, the bioaccumulation factor (BAF) was further analyzed. BAF is defined as the ratio of the mercury concentration in brown rice to the total mercury concentration in soil, calculated as:

$$\text{BAF} = (\text{Pollutant content in rice plant})/(\text{Pollutant content in soil}) \quad (1)$$

In this study, the bioaccumulation factors for total mercury and methylmercury in brown rice are denoted as BAF_{Hg} and BAF_{MeHg}, respectively, where concentrations in both soil and plant tissues are expressed in ng g^{-1}.

All quantitative data are expressed as mean ± standard deviation (SD). Experimental data were processed using Microsoft Office Excel. IBM SPSS Statistics (v22.0.0.0) was used for statistical analysis, including Levene's test for homogeneity of variance ($p > 0.05$), Spearman's rank correlation, one-way and two-way analysis of variance (ANOVA), and analysis of covariance (ANCOVA). Duncan's multiple range test was used for post hoc comparisons. The significance level was set at $p < 0.05$. Figures and replicate-level linear regressions were prepared using Origin (v2025b) software.

Random Forest analysis was conducted to identify key environmental drivers, with variable importance evaluated using the percentage increase in mean squared error (%IncMSE). The analysis was performed using the randomForest (v 4.7-1.1) package in R (v 4.4.1) with ntree = 1000 and 5-fold cross-validation to determine the optimal mtry. The optimal mtry was 9 for BAF_{Hg} and 2 for BAF_{MeHg}, selected based on the smallest root mean square error (RMSE) from cross-validation. The models explained 83.62% of the variance in BAF_{Hg} and 43.32% of the variance in BAF_{MeHg}. Panicle number per plant was used as an indicator of rice growth status.

Statistical Code Development: The R code for implementing the random forest analysis was initially drafted with the assistance of a generative artificial intelligence tool (DeepSeek, OpenAI by deepseek.ai (https://deepseek.ai/, accessed on 23 January 2026)) to accelerate the coding process. All AI-generated code was critically reviewed, verified for correctness, and extensively modified by the authors to ensure it accurately implemented the intended

statistical models as described in the main text. The authors take full responsibility for the final code, its execution, and the interpretation of all statistical results.

3. Results

3.1. Soil Aging Effect on Hg Uptake

By comparing Hg accumulation in rice grown in freshly processed versus aged pot soil, we observed that, in both years of pot experiments (total Hg remained stable; Hg speciation was not assessed), brown rice Hg concentrations were significantly greater than those in the field (Figure 1A). Mean mercury concentrations in brown rice were 39.67 ± 7.31 ng g^{-1}, 29.72 ± 6.82 ng g^{-1}, and 11.88 ± 3.83 ng g^{-1} (n = 33) for the 2021 pot, 2022 pot, and 2022 Field B trials, respectively. The 2021 pot value was significantly greater than both the 2022 pot and 2022 Field B values (Figure 1A). The effect sizes (Cohen's d) were 1.41 for the comparison between 2021 and 2022 pots, and 4.76 for the comparison between 2021 pots and 2022 Field B, indicating large differences in both cases.

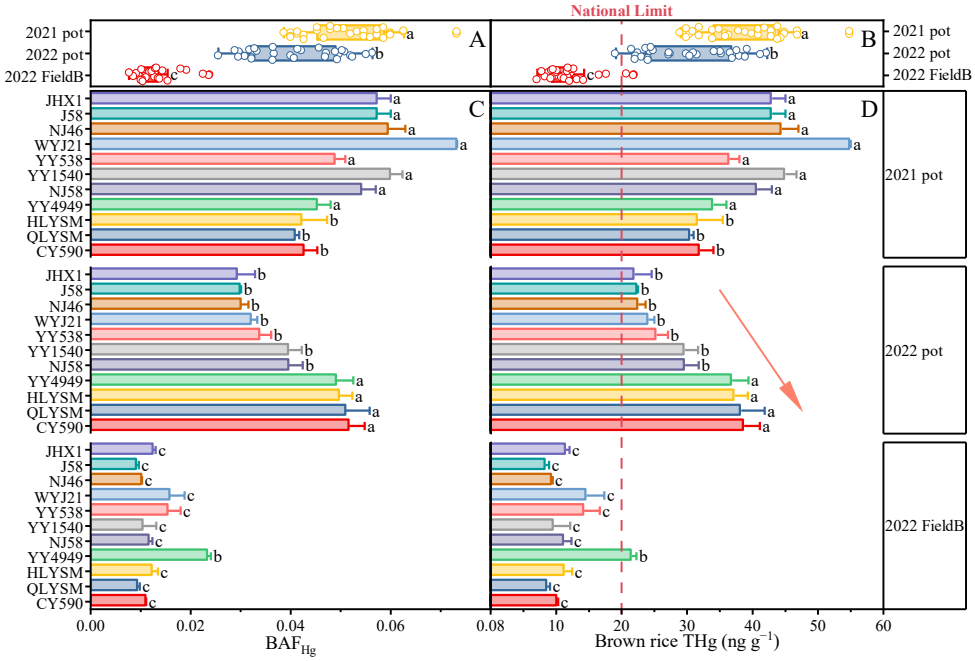


Figure 1. Average and specific rice cultivar bioaccumulation factor (BAF) for brown rice Hg (**A**,**C**) and Hg concentration (**B**,**D**) across different soils. Data within each panel are arranged in ascending order of their mean values (indicated by arrows). Error bars represent standard deviation (SD) (n = 3). Different lowercase letters above bars denote significant differences ($p < 0.05$) between years for the same rice cultivar.

In 2022, field-grown rice showed significantly lower mercury concentration and BAF_{Hg} than pot-grown rice from 2021 to 2022. While most cultivars exhibited a significant increase in mercury accumulation in the second year of pot trials, some decreased. For example, YY4949 ranked highest in 2022 Field B but only intermediate in the 2022 pot trial, whereas QLYSM, the lowest accumulator in the field, became a high accumulator in pots. Almost all pot-grown cultivars exceeded the national food safety standard (20 ng g^{-1}) in both the 2021 and 2022 pot trials, while in the field, only YY4949 surpassed this limit.

To further examine the stability of cultivar rankings under soil aging and their consistency with field conditions, Spearman's rank correlation coefficients were calculated for the 11-cultivar set (Table S6). The correlation between 2021 and 2022 pot rankings was negative and significant ($\rho = -0.755$, $p = 0.007$), indicating that soil aging substantially altered the relative order of cultivars in pots. When comparing pot rankings with field

rankings (2022 Field B), the correlations were weak and non-significant for both 2021 pots (ρ = 0.073, p = 0.832) and 2022 pots (ρ = 0.000, p = 1.000).

For a second set of 10 cultivars, Spearman's rank correlations were calculated between pot and field rankings (Table S7). The correlation between pot rankings and those at Field A was weak and non-significant (ρ = −0.152, p = 0.676), as was the correlation between pot rankings and those at Field B (ρ = 0.406, p = 0.244).

3.2. Pot-Field Hg Accumulation Validation

By contrasting Hg accumulation in the same cultivars grown in pots and at two field sites, we found that pot-grown brown rice had significantly higher Hg concentrations than field-grown rice (Figure 2B). The mercury concentration in pot-grown brown rice (39.06 ± 10.02 ng g^{-1}) was significantly greater than those at both field sites (Field A: 7.42 ± 1.56 ng g^{-1}; Field B: 17.55 ± 7.37 ng g^{-1}), being 5.26 and 2.23 times that of Field A and Field B, respectively (Figure 2B). Comparing the 10 cultivars between 2022 pot and the two field sites (Field A and Field B), the correlations were also weak (Field A: ρ = −0.152, p = 0.676; Field B: ρ = 0.406, p = 0.244; Table S7).

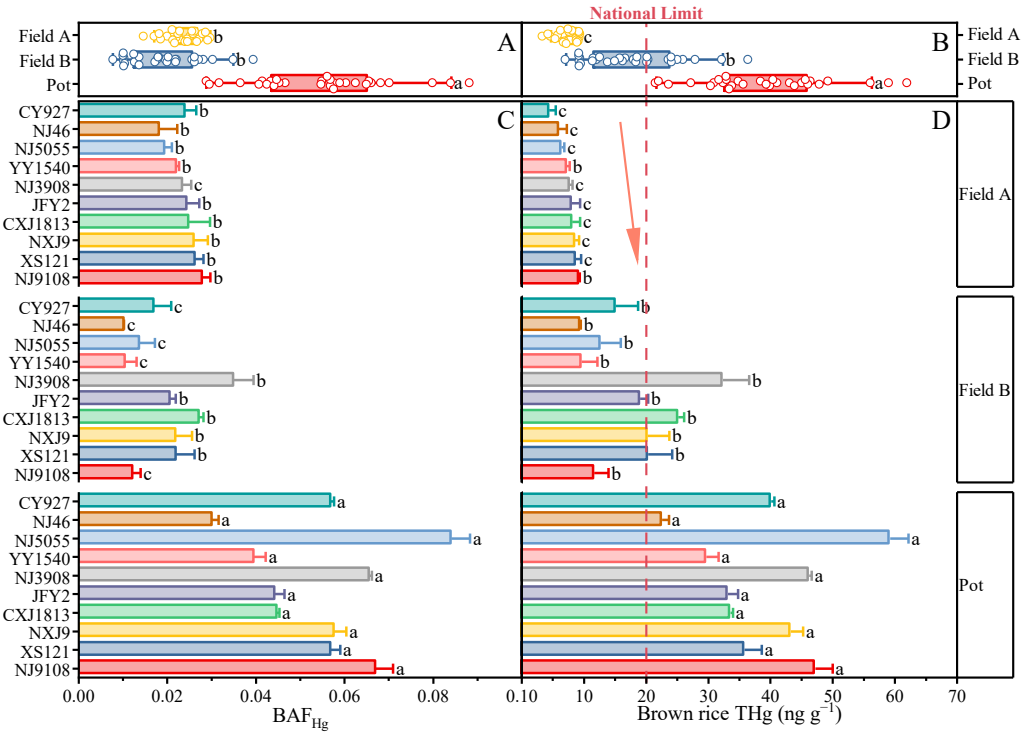


Figure 2. Average and specific rice cultivar bioaccumulation factor (BAF) for brown rice Hg (**A**,**C**) and Hg concentration (**B**,**D**) across different growing environments (2022). Data within each panel are arranged in ascending order of their mean values (indicated by arrows). Error bars represent standard deviation (SD) (n = 3). Different lowercase letters above bars denote significant differences (p < 0.05) between growing environments for the same rice cultivar.

Both BAF_{Hg} and brown rice mercury concentration were significantly greater in the pot system than in Field A or Field B (Figure 2C,D). However, the trends for brown rice mercury concentration and BAF_{Hg} differed across soils with varying contamination levels. Specifically, while mercury concentrations in Field B were significantly lower than those in pots but greater than those in Field A, the BAF_{Hg} values in Field A and Field B were statistically similar and both significantly lower than that of the pot treatment (Figure 2A,B). For example, in cultivar NJ5055, the brown rice mercury concentration in Field B was 98.85% greater than in Field A (p < 0.05) but 78.63% lower than in pots (p < 0.05). Conversely, its

BAF_{Hg} in Field B was 29.31% lower than in Field A ($p < 0.05$) and 83.76% lower than in pots ($p < 0.05$).

To evaluate the predictive relationship between pot and field Hg concentrations, replicate-level linear regressions were performed between brown rice Hg concentrations in pots and those at the two field sites (Figure 3). For Field A, the regression equation was y = 0.0276x + 6.54 (R^2 = 0.040, p = 0.312, r = 0.201, n = 30). For Field B, the equation was y = 0.1404x + 12.38 (R^2 = 0.034, p = 0.324, r = 0.186, n = 30). A replicate-level regression was also performed between brown rice Hg concentrations at Field A and Field B (Figure 4). The regression equation was y = 1.29x + 8.06 (R^2 = 0.055, p = 0.211, r = 0.235, n = 30). Similar replicate-level regressions were performed using the BAF_{Hg} to account for differences in soil Hg concentrations. Between pot and field BAF values, the relationships were negligible (Figure S2): Field A (y = 0.0381x + 0.0215, R^2 = 0.024, p = 0.417, r = 0.154, n = 30) and Field B (y = 0.0978x + 0.0141, R^2 = 0.031, p = 0.349, r = 0.177, n = 30). Between BAF values at Field A and Field B, the relationship was weak (y = 0.75x + 0.0017, R^2 = 0.112, p = 0.070, r = 0.335, n = 30) (Figure S3).

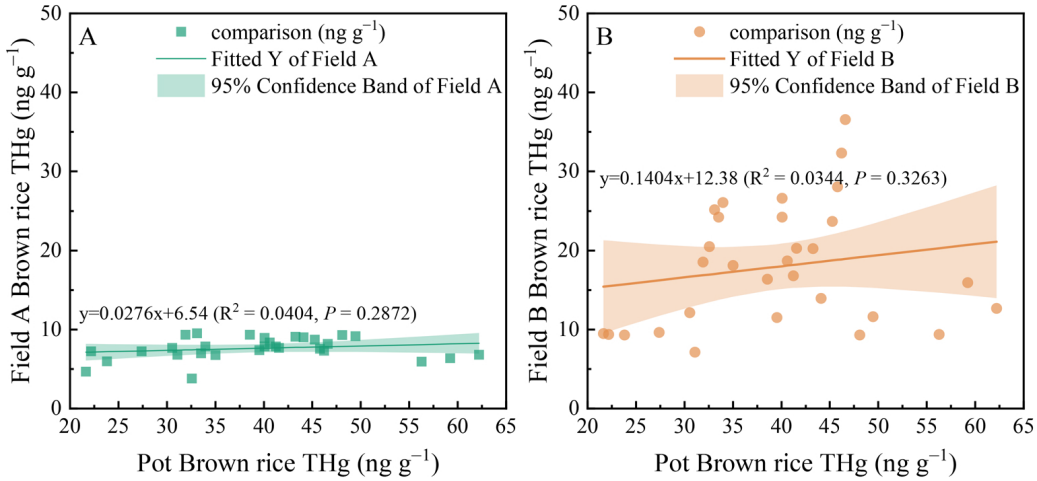


Figure 3. Scatter plots of brown rice total mercury (THg) concentrations (ng g^{-1}) in pot experiments (soil Hg: 0.70 mg kg^{-1}) versus those at two field sites: (**A**) Field A (low-Hg soil, 0.33 mg kg^{-1}) and (**B**) Field B (high-Hg soil, 0.93 mg kg^{-1}). Each point represents an individual replicate (n = 3 per cultivar, 10 cultivars, 30 points per site). Solid lines indicate linear regression fits, and shaded areas represent the 95% confidence intervals. Regression results: (**A**) y = 0.0276x + 6.54, R^2 = 0.040, p = 0.312, Pearson's r = 0.201; (**B**) y = 0.1404x + 12.38, R^2 = 0.034, p = 0.324, Pearson's r = 0.186. The near-zero slopes and low R^2 values indicate that pot Hg concentrations explain less than 5% of the variation in field Hg concentrations at both sites, confirming that pot-based data showed limited predictive capacity under the tested conditions.

Beyond using BAF to account for soil Hg concentration, analysis of covariance (ANCOVA) was performed with field site as the fixed factor and soil pH and OM as covariates. The overall model was significant (F = 17.56, $p < 0.001$, adjusted R^2 = 0.457), explaining 48.5% of the variance in brown rice Hg concentration. After controlling for pH and OM, the effect of site remained significant (F = 5.83, p = 0.019, η^2 = 0.094), with Field A exhibiting a lower estimated Hg concentration (7.04 ng g^{-1}) than Field B (18.51 ng g^{-1}). The covariates pH (F = 0.40, p = 0.529) and OM (F = 0.50, p = 0.481) did not show significant independent effects. It should be noted that the Levene's test indicated heterogeneity of variances across groups ($p < 0.001$), and the lack-of-fit test was significant (p = 0.001), suggesting that the relationship between the covariates and Hg concentration may not be strictly linear. Nevertheless, the significant site effect remained robust under alternative model specifications.

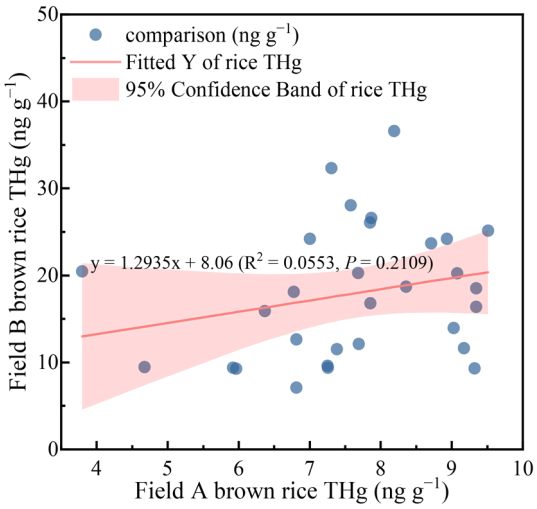


Figure 4. Scatter plot of brown rice total mercury (THg) concentrations (ng g^{-1}) between two field sites: Field A (low-Hg soil, 0.33 mg kg^{-1}) and Field B (high-Hg soil, 0.93 mg kg^{-1}). Each point represents an individual replicate (n = 3 per cultivar, 10 cultivars, 30 points per site). The solid line indicates linear regressions fit, and the shaded area represents the 95% confidence interval. Regression results: y = 1.29x + 8.06, R^2 = 0.0553, p = 0.2109, Pearson's r = 0.2352 (n = 30). The near-zero slopes and R^2 values below 0.05 indicate that pot Hg concentrations explain less than 5% of the variation in field Hg concentrations, confirming that pot-based data showed limited predictive capacity under the tested conditions.

3.3. Microenvironmental Controls on Hg Accumulation

To investigate the effect of the pot environment on mercury accumulation in brown rice, we conducted a buried-pot experiment. This experiment was designed to isolate the effects of the root-zone micro-environment (e.g., temperature, light) by burying pots directly in the field (BP treatment). During the tillering stage, mean soil temperature in GP was 36.5 °C—approximately 6–8 °C higher than in BP—while light intensity was reduced by more than 90% (Table S8).

For the three tested rice cultivars, BAF_{MeHg}, Hg, and MeHg concentration in brown rice were significantly greater in the GP treatment than in the other experimental groups (Figures 5A, 6A and S4A), though no significant differences were observed among the cultivars themselves (Figure S5). The GP treatment also resulted in markedly greater levels of these three parameters compared to other cultivation environments (Figures 5B, 6B and S4B). Cultivar NJ3908 exhibited the highest BAF_{MeHg} under the GP treatment, which was significantly greater than that of other cultivars and treatments (Figure 5A). Across all treatments, the mean concentrations of total mercury and methylmercury in brown rice followed a consistent descending order: GP > BP > BR > CS (Figures 6B and S4B).

To identify the key drivers of the Hg accumulation difference between the BP and GP treatments, a Random Forest analysis (%IncMSE) was performed on their climatic, edaphic, and plant-trait variables (Figure 7). The model explained 83.62% of the variance in BAF_{Hg} and 43.32% of the variance in BAF_{MeHg}, with 5-fold cross-validation confirming model stability. For the BAF_{Hg}, soil temperature (10.55, $p < 0.05$) and light intensity (10.46, $p < 0.01$) were the primary predictors, followed by air humidity and air temperature. For the BAF_{MeHg}, rice subspecies (9.24, $p < 0.05$) and soil temperature (8.12, $p < 0.05$) were most influential, with light intensity (7.15, $p < 0.01$) also contributing significantly. Soil total Hg concentration showed no explanatory power (%IncMSE = 0, ns).

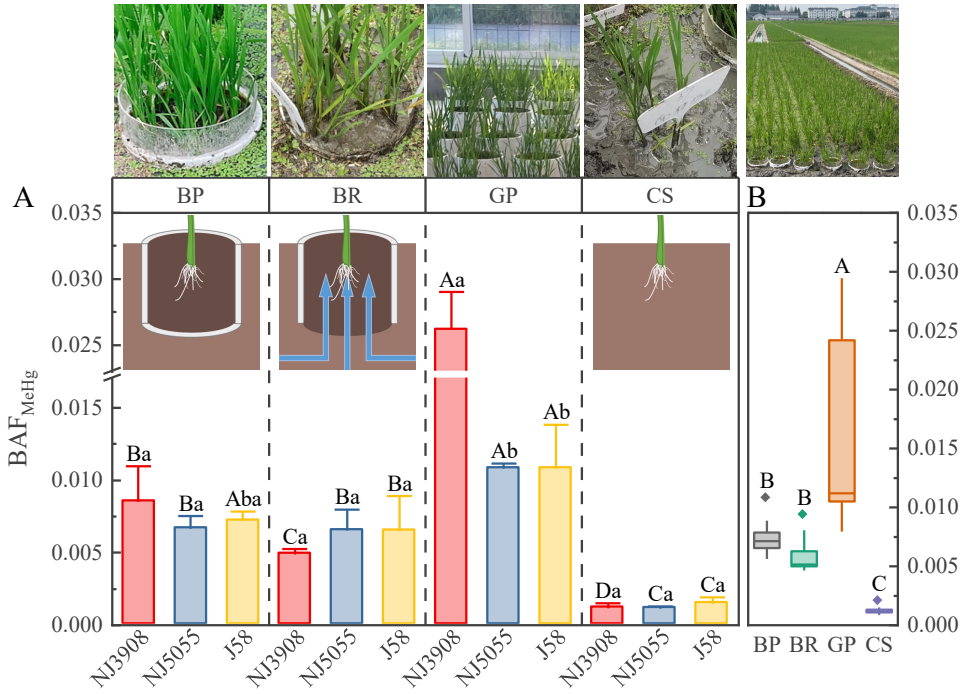


Figure 5. The average (**B**) and specific rice cultivar (**A**) methylmercury bioaccumulation factor (BAF_{MeHg}) in brown rice under different cultivation treatments. Error bars represent standard deviation (SD; n = 3). Different lowercase letters above bars indicate significant differences ($p < 0.05$) among cultivars within the same treatment. Uppercase letters indicate significant differences ($p < 0.05$) among treatments for the same cultivar. Abbreviations: BP, Buried Pot; BR, Block-flowing PVC ring; GP, Greenhouse Pot; CS, Clean Soil.

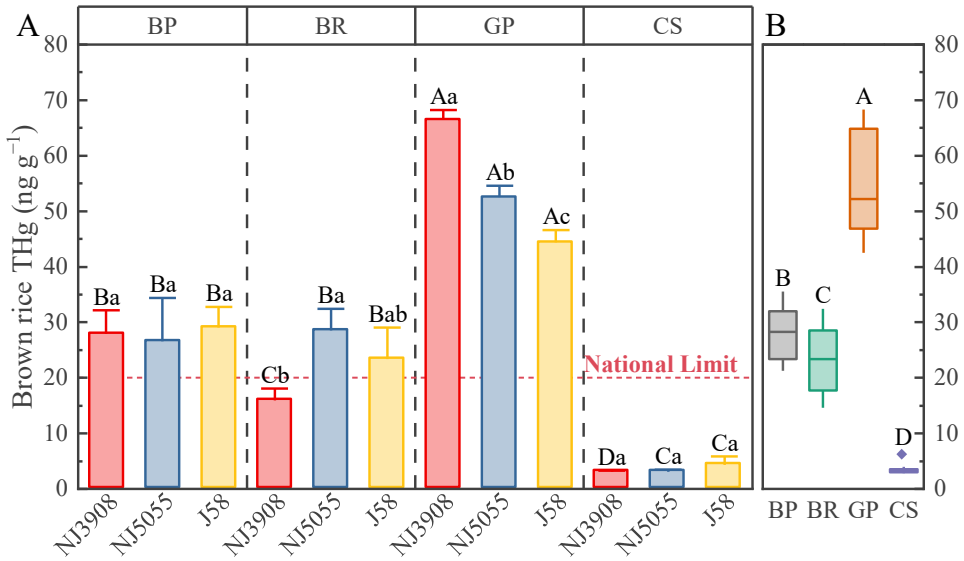


Figure 6. Average (**B**) and specific rice cultivar (**A**) total mercury (Hg) concentration in brown rice under different cultivation treatments. Error bars represent standard deviation (SD; n = 3). Different lowercase letters above bars indicate significant differences ($p < 0.05$) among cultivars within the same treatment. Uppercase letters indicate significant differences ($p < 0.05$) among treatments for the same cultivar. Abbreviations: BP, Buried Pot; BR, Block-flowing PVC ring; GP, Greenhouse Pot; CS, Clean Soil.

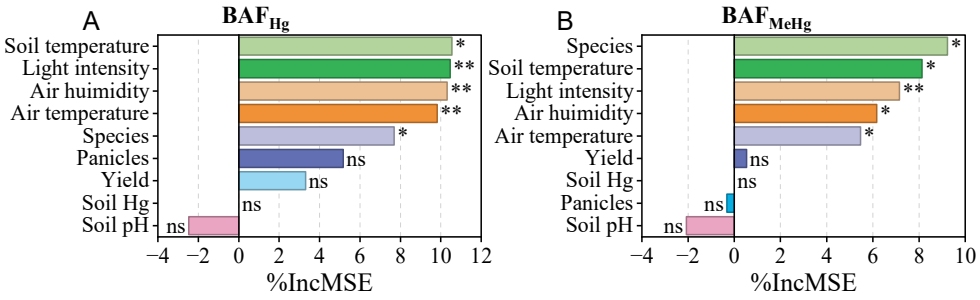


Figure 7. Relative importance of predictors (%IncMSE) from the random forest model for bioaccumulation factors of total mercury (BAF_{Hg}) (**A**) and methylmercury (BAF_{MeHg}) (**B**) using data of the Buried Pot (BP) and Greenhouse Pot (GP) treatments. *, **, and ns indicate statistical significance at $p = 0.05$, $p = 0.01$, and no significance ($p > 0.05$), respectively.

4. Discussion

4.1. Pot Experiments Show Limited Predictive Transferability for Field Mercury Accumulation in Field Grown Rice

A central aim of this study was to assess whether pot-based screening reliably predicts field-level Hg accumulation in rice. Across multiple cultivars, years, and sites, we consistently observed that pot-grown rice accumulated significantly higher Hg concentrations in brown rice than field-grown rice, with pot values ranging from 2.2 to 5.3 times those measured at two field sites (Figures 1 and 2). More importantly, cultivar rankings derived from pot experiments showed weak and non-significant correlations with field rankings (Spearman's $\varrho = 0.073$–0.406, all $p > 0.05$), and replicate-level regressions revealed that pot Hg concentrations explained less than 5% of the variation in field Hg concentrations ($R^2 < 0.05$, Figure 3). Furthermore, a weak and non-significant relationship was also observed between the two field sites themselves (Figure 4), indicating that even under field conditions, cultivar-specific Hg accumulation patterns are not readily transferable across locations. These findings demonstrate that, under the tested conditions, pot experiments not only overestimate the magnitude of Hg accumulation but also fail to preserve the relative order of cultivar performance.

The systematic overestimation observed here is consistent with previous reports from the Yangtze River Delta, where a field survey reported brown rice Hg concentrations ranging from 0.30–39.8 ng g^{-1} Hg in brown rice (mean 4.53 ng g^{-1}) [23], while a parallel pot experiment using soils from the same region yielded substantially higher values (mean 20.2 ng g^{-1} in unspiked control soil) [24]. This discrepancy cannot be fully attributed to differences in soil Hg concentrations, as the bioaccumulation factor (BAF) also exhibited negligible correlation between pot and field systems ($R^2 < 0.05$; Figure S2). Rather, the overestimation likely arises from a combination of factors unique to the pot environment, including elevated root-zone temperature, attenuated light intensity, and physical soil disturbance—all of which can enhance Hg bioavailability and methylation while suppressing photodegradation [25–31].

The lack of transferable cultivar rankings across environments is equally noteworthy. Even between the two field sites (Field A and Field B), cultivar-specific Hg accumulation patterns were not consistent (Figure 4), with site explaining a significant portion of the variation even after accounting for soil pH and organic matter (ANCOVA, $F = 5.83$, $p = 0.019$). This context-dependent performance aligns with previous studies showing that environmental factors—such as climate, soil properties, and agricultural practices—can explain up to 40.6% of grain Hg variation [32], often overshadowing genotypic differences. Thus, while pot experiments may serve as a useful first step for initial cultivar screening, their predictive capacity for field performance is inherently limited under the tested conditions.

Together, these findings highlight two important implications. First, the pot environment creates a "worst-case" scenario that systematically elevates Hg accumulation, which may lead to overestimation of food safety risks if extrapolated directly to the field. Second, the lack of consistent cultivar rankings across environments underscores the necessity of multi-site, multi-year field validation for region-specific cultivar recommendations. Pot experiments remain valuable for controlled mechanistic studies (e.g., exploring Hg uptake pathways or microbial processes), but their use for predictive risk assessment or cultivar selection should be accompanied by rigorous field verification.

4.2. Microenvironmental Drivers of Mercury Overestimation in Pot Systems

To disentangle the effects of the pot microenvironment from other confounding variables, we introduced a buried-pot (BP) treatment that mimicked field root-zone conditions while retaining the same soil and pot geometry as the conventional greenhouse pot (GP) treatment. Random forest analysis identified elevated soil temperature and reduced light intensity as the primary factors driving the higher Hg and MeHg accumulation in GP relative to BP (Figure 7). Deposition fluxes were comparable across all outdoor sampling locations (Figure S1 and Table S3) and significantly lower than those in the glass greenhouse (GP), where deposition was negligible ($p < 0.05$), indicating that differences in atmospheric input did not explain the observed variation.

The microclimatic divergence between GP and BP was substantial (Table S8). These artificial conditions likely promote Hg accumulation through two interconnected pathways. First, the elevated root-zone temperatures enhance microbial Hg methylation activity [33] and increase the bioavailability of soil Hg [25–27]; they also stimulate elemental Hg (Hg^0) volatilization and subsequent foliar absorption [34–38]. Second, the marked attenuation of UV radiation under greenhouse glazing slows the photodegradation of methylmercury in paddy water and the rhizosphere, thereby reducing a key natural sink for MeHg [30,31]. In addition, the physical handling of soil—including sieving, air-drying, and short-term aging—maintains Hg in more bioavailable forms than in undisturbed field soil [39,40], and the restricted root volume in conventional pots may further amplify uptake [41].

These findings collectively suggest that the pot environment creates a "worst-case" microclimate that systematically elevates Hg accumulation relative to field conditions. The overestimation is not merely a matter of scale but reflects fundamental alterations in temperature, light, and soil integrity. Consequently, extrapolating absolute Hg concentrations from pot experiments to field risk assessments should be approached with caution. Nevertheless, such controlled systems remain valuable for elucidating mechanistic pathways—for example, isolating the effects of temperature or light on Hg methylation and demethylation—provided their limitations are explicitly acknowledged.

4.3. Region-Specific Performance of Low-Hg Cultivars and Implications for Screening

A key practical question arising from the observed pot–field discrepancies is whether low-Hg cultivars identified in pots perform consistently across different field environments. Our results clearly indicate that they do not. Cultivar rankings in pots showed no correlation with those at either field site (Figure 3), and more importantly, rankings were also inconsistent between the two field sites themselves (Figure 4). For instance, NJ3908 exhibited high Hg accumulation in pots and at Field B but not at Field A, whereas NJ9108 showed the opposite pattern (Figure 2). These site-dependent cultivar differences are quantitatively supported by the regression analyses: the low R^2 values and weak correlation coefficients (r = 0.201 for Field A and r = 0.186 for Field B) between pot and field Hg concentrations (Figure 3) indicate no predictive relationship, while the weak correlation

between the two field sites (Figure 4) further demonstrates that cultivar performance is strongly context-dependent.

To further explore whether soil properties account for this site-dependent performance, we conducted an ANCOVA with site as the fixed factor and soil pH and organic matter (OM) as covariates. After adjusting for these soil properties, the difference between Field A and Field B remained statistically significant ($F = 5.83$, $p = 0.019$). This finding was further supported by BAF-based analyses (Figures S2 and S3), which normalized for soil total Hg and showed weak and non-significant correlations between field sites ($R^2 = 0.112$, $p = 0.070$; Figure S3). Together, these results indicate that environmental factors beyond soil pH and OM—and beyond soil Hg concentration alone—govern cultivar-specific Hg accumulation patterns.

The observed site-dependent performance likely reflects a combination of environmental factors beyond the measured soil properties. Mercury in paddy systems originates from both anthropogenic soil contamination and atmospheric deposition [42], and its accumulation in rice can be influenced by regional industrial activities [43], precipitation regimes [44], and local farming practices [45]. Moreover, microbial-mediated mercury methylation—a central process in paddies [15,46,47], also varies with local soil and climatic conditions [48]. Added to this are cultivar-specific traits, such as root exudates [33] and iron-plaque formation capacity [49], which further influence mercury uptake. Together, these processes operate at multiple regulatory nodes, underpinning the region-dependent performance of rice cultivars. Thus, pot-based data cannot reliably predict field accumulation or support broad cultivar screening. Low-Hg rice cultivar selection must be region-specific to avoid systematic environmental bias.

This study was conducted in a single rice-growing region with a limited number of cultivars and a limited range of soil Hg concentrations; the conventional pot size may have introduced root confinement that could amplify the overestimation. While our results demonstrate systematic overestimation under the tested conditions, we acknowledge that the magnitude of pot–field discrepancies may vary with soil properties, rice genotypes, and experimental designs. In certain contexts—such as controlled mechanism studies or initial germplasm screening—pot experiments may still offer valuable insights, provided their limitations are explicitly considered. However, based on the influence of local microclimate on Hg accumulation observed here, cultivar selection should target relatively small geographic areas—i.e., deployed in regions similar to where they were screened. Pot experiments may serve for initial selection but should be validated in at least two representative field sites. Despite these considerations, the evidence suggests that pot experiments can overestimate Hg accumulation and that low-Hg cultivar screening is likely region-specific.

5. Conclusions

Through systematic analysis of multi-year data from pot and parallel field experiments, this study provides evidence that, under the tested conditions, the pot environment overestimates the potential for mercury accumulation in rice and shows limited predictive capacity for field-level cultivar performance. Specifically: (1) Brown rice mercury content in pots is significantly greater than that in the field, with replicate-level regressions showing that pot Hg concentrations explain less than 5% of the variation in field Hg concentrations across sites. The use of aged soil cannot replicate the patterns of field rice mercury accumulation. (2) Soil temperature and light intensity were the core environmental factors driving this difference. (3) Cultivar performance varied across field sites, with no consistent ranking between locations, indicating that low-Hg cultivar screening is region-specific.

Despite these limitations, pot experiments remain a valuable tool for controlled mechanistic studies—for example, investigating Hg uptake pathways, microbial methylation

processes, or the effects of soil amendments—where their high controllability offers distinct advantages. However, their use for predictive risk assessment or cultivar selection requires careful interpretation and should be complemented by field validation. Given that this study was conducted in a single rice-growing region with a limited set of cultivars and a limited range of soil contamination levels, we strongly recommend caution in interpreting pot experiments for cultivar screening or risk assessment for Hg, as it is prone to misleading conclusions. Multi-site, multi-year field validation should be prioritized to ensure accurate risk assessment and region-specific cultivar recommendation. Future research could explore semi-field mesocosm systems as an intermediate between pots and field conditions, offering more realistic root environments with some experimental control.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/agriculture16080907/s1, Figure S1: Schematic diagram showing the locations of atmospheric deposition collectors (DEP) and the pot experiment treatments. Abbreviations: BP, Buried Pot; BR, Block-flowing PVC ring; GP, Greenhouse Pot; CS, Clean Soil; Figure S2: Scatter plots of the bioaccumulation factor (BAF) for total mercury (THg) in brown rice between pot experiments (soil Hg: 0.70 mg kg^{-1}) and two field sites: (A) Field A (low-Hg soil, 0.33 mg kg^{-1}) and (B) Field B (high-Hg soil, 0.93 mg kg^{-1}). Each point represents an individual replicate (n = 3 per cultivar, 10 cultivars, 30 points per site). Solid lines indicate linear regression fits, and shaded areas represent the 95% confidence intervals. Regression results: (A) y = 0.0381x + 0.0215, R^2 = 0.024, p = 0.417, Pearson's r = 0.1539; (B) y = 0.0978x + 0.0141, R^2 = 0.031, p = 0.349, Pearson's r = 0.1770; Figure S3: Scatter plot of the bioaccumulation factor (BAF) for total mercury (THg) in brown rice between Field A (low-Hg soil, 0.33 mg kg^{-1}) and Field B (high-Hg soil, 0.93 mg kg^{-1}). Each point represents an individual replicate (n = 3 per cultivar, 10 cultivars, 30 points total). The solid line indicates the linear regression fit, and the shaded area represents the 95% confidence interval. Regression results: y = 0.75x + 0.0017, R^2 = 0.1124, p = 0.07, r = 0.3353 (n = 30); Figure S4: Methylmercury (MeHg) concentration in brown rice under different cultivation treatments, showing (A) a comparison across cultivars and (B) the average effect across all cultivars. Different lowercase letters above bars indicate significant differences ($p < 0.05$) among cultivars within the same treatment. Uppercase letters indicate significant differences ($p < 0.05$) among treatments for the same cultivar. Abbreviations: BP, Buried Pot; BR, Block-flowing PVC ring; GP, Greenhouse Pot; CS, Clean Soil; Figure S5. Comparison of (a) total mercury (Hg) concentration, (b) methylmercury (MeHg) concentration, (c) total Hg bioaccumulation factor (BAF_{Hg}), and (d) MeHg bioaccumulation factor (BAF_{MeHg}) among different rice cultivars across all cultivation treatments; Table S1: Characteristics of the rice cultivars used in Experiment I (mean ± SD; n = 3); Table S2: Characteristics of the rice cultivars used in Experiment III (mean ± SD; n = 3); Table S3: Mercury concentrations in atmospheric dry and wet deposition during the pot experiment period; Table S4: Application rates of the water-soluble fertilizer used in the pot experiments; Table S5: Method detection limits and recoveries of standard reference materials for mercury determination; Table S6. Spearman's rank correlation coefficients among cultivar rankings under different conditions (11-cultivar set, n = 11); Table S7. Spearman's rank correlation coefficients among cultivar rankings under different conditions (10-cultivar set, n = 10); Table S8: Key microclimate parameters for the Buried Pot (BP) and Greenhouse Pot (GP) treatments during a representative monitoring period (tillering stage, August 2023).

Author Contributions: Conceptualization, Z.D. and J.D.; methodology, J.D. and L.Z.; software, L.Z. and Z.Z.; validation, X.M., F.L. and Y.G.; formal analysis, X.A.; investigation, F.L. and Y.G.; resources, X.L.; data curation, Z.S.; writing—original draft preparation, L.Z.; writing—review and editing, J.D.; visualization, L.Z.; supervision, Z.D. and X.L.; project administration, Z.D. and L.Z.; funding acquisition, Z.D. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: The authors declare no conflicts of interest.

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